Fibonacci heaps exercises

19-E01 Extract-min root list

Difficulty: Basic

The heap has exactly nodes2,7,5,with leaf children7,5 under root2. Give temporary roots after extract-min,then consolidation and new minimum.

19-E02 Cascading cut

Difficulty: Advanced

Lowering x violates heap order. Parent p is marked and grandparent g unmarked. Trace cuts and mark changes.

19-E03 Minimum subtree by degree

Difficulty: Research

With F0=0,F1=1,F2=1,use $\text{size} \geq F_{k+2}$ to bound size at degree5. Why is this a Fibonacci heap?

19-E04 Total cascade account

Difficulty: Research

A decrease-key cuts c≥1 nodes. Bound Δ(t+2m) using roots t and marks m. If actual cost≤ac+b,choose a scaling constant explicitly to prove O(1) amortized cost. Must the first cut node have been marked?

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19-E05 A binary degree-array bound is insufficient

Difficulty: Research

Insert0..16 and extract0,giving a16-node degree-four tree rooted at1. Lower id16 to−1 and extract it. id16 was a leaf child of id15. Give root1’ s degree and total nodes;must A[0..floor($\log_2$n)] suffice?

19-E06 Erase a specified handle among equal minima

Difficulty: Advanced

All int values and duplicate keys are permitted. Why may assigning x.key=INT_MIN and extracting minimum delete the wrong identity? Give erase(handle) without key modification and its amortized cost.

> Complete your derivations before opening the separate solutions PDF.

Fibonacci heaps exercises (continued)

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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